Analysis of JavaScript Web Applications Using SAFE 2.0

ICSE Demo 2017

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PLRG @ KAIST

JavaScript Web Application



JavaScript Web Application



Error





Analyzer

JavaScript Web Application



Error





Analyzer

JavaScript Web Application



Error





Analyzer

JavaScript Web Application



Error





Analyzer

Use by Analysis Developers

- Poor **usability** for **analysis developers**

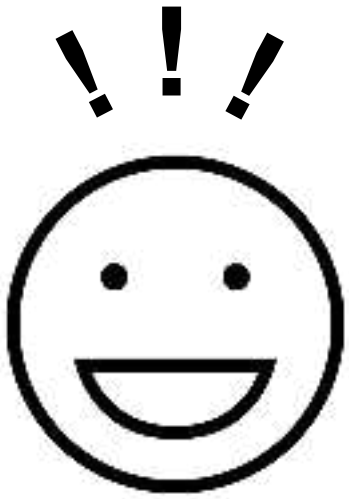


Analysis Developers

- **configure** analysis techniques
- **add** new techniques
- **debug** their implementation

SAFE 2.0

- Advanced version of **SAFE**
- **SAFE 2.0** with **three main features**



Analysis Developers

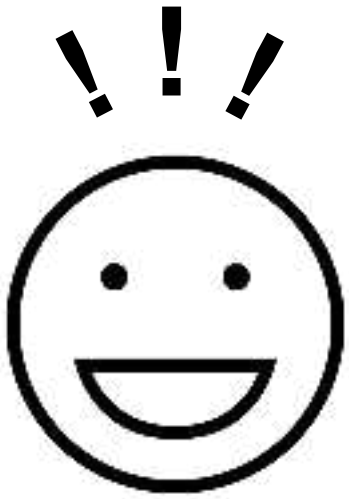
Pluggability

Extensibility

Debuggability

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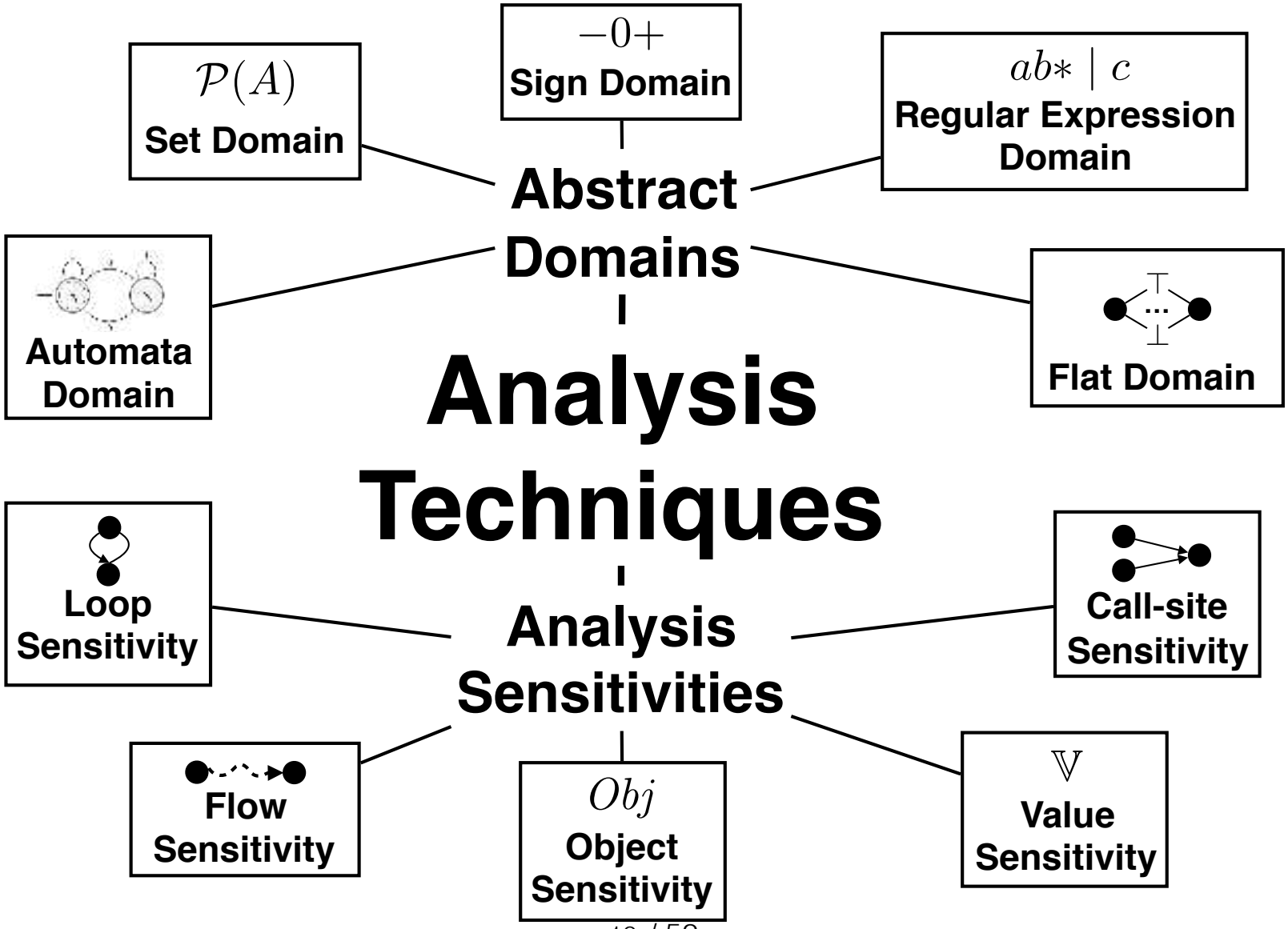
Analysis Developers

Pluggability

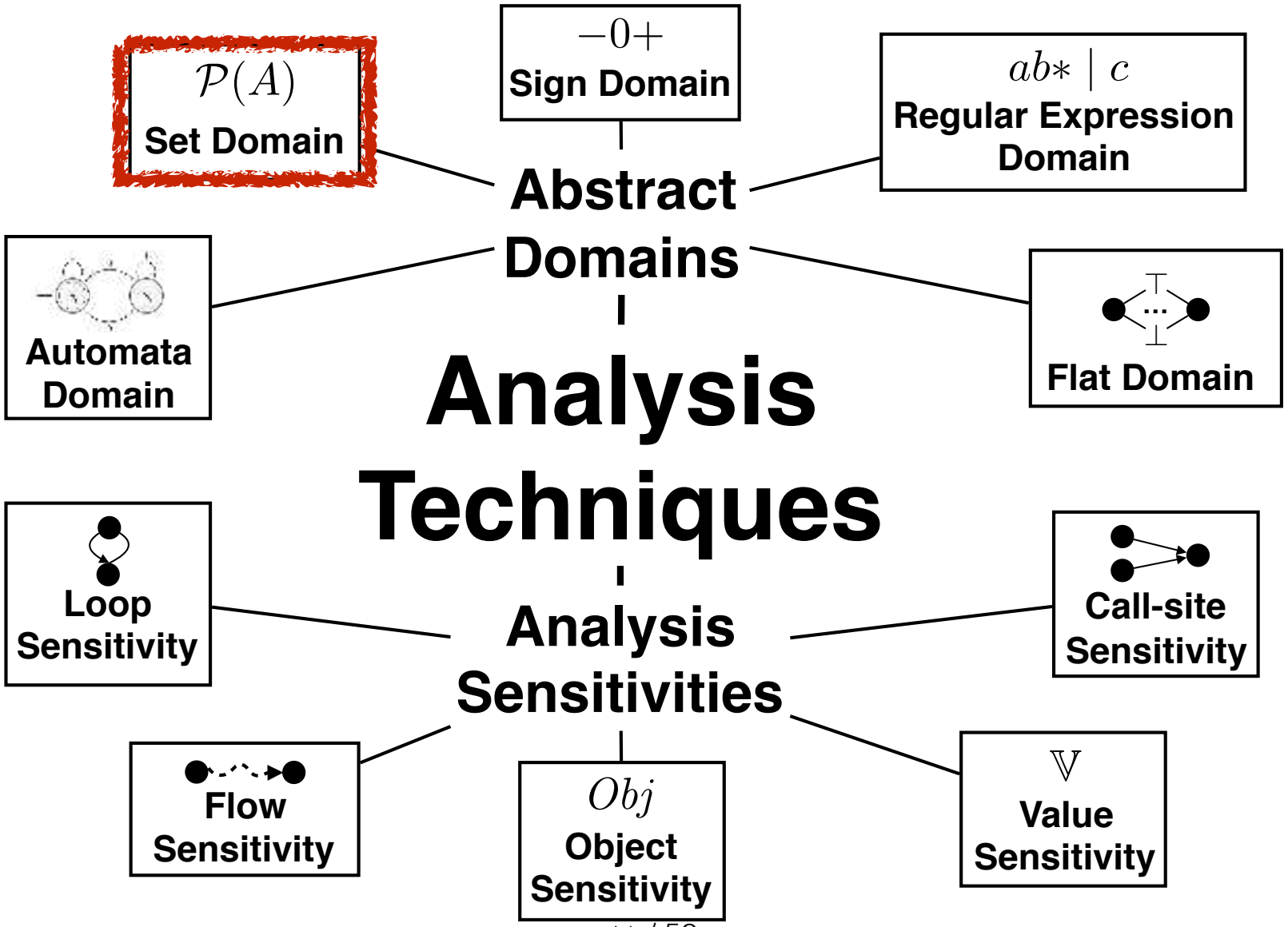
Extensibility

Debuggability

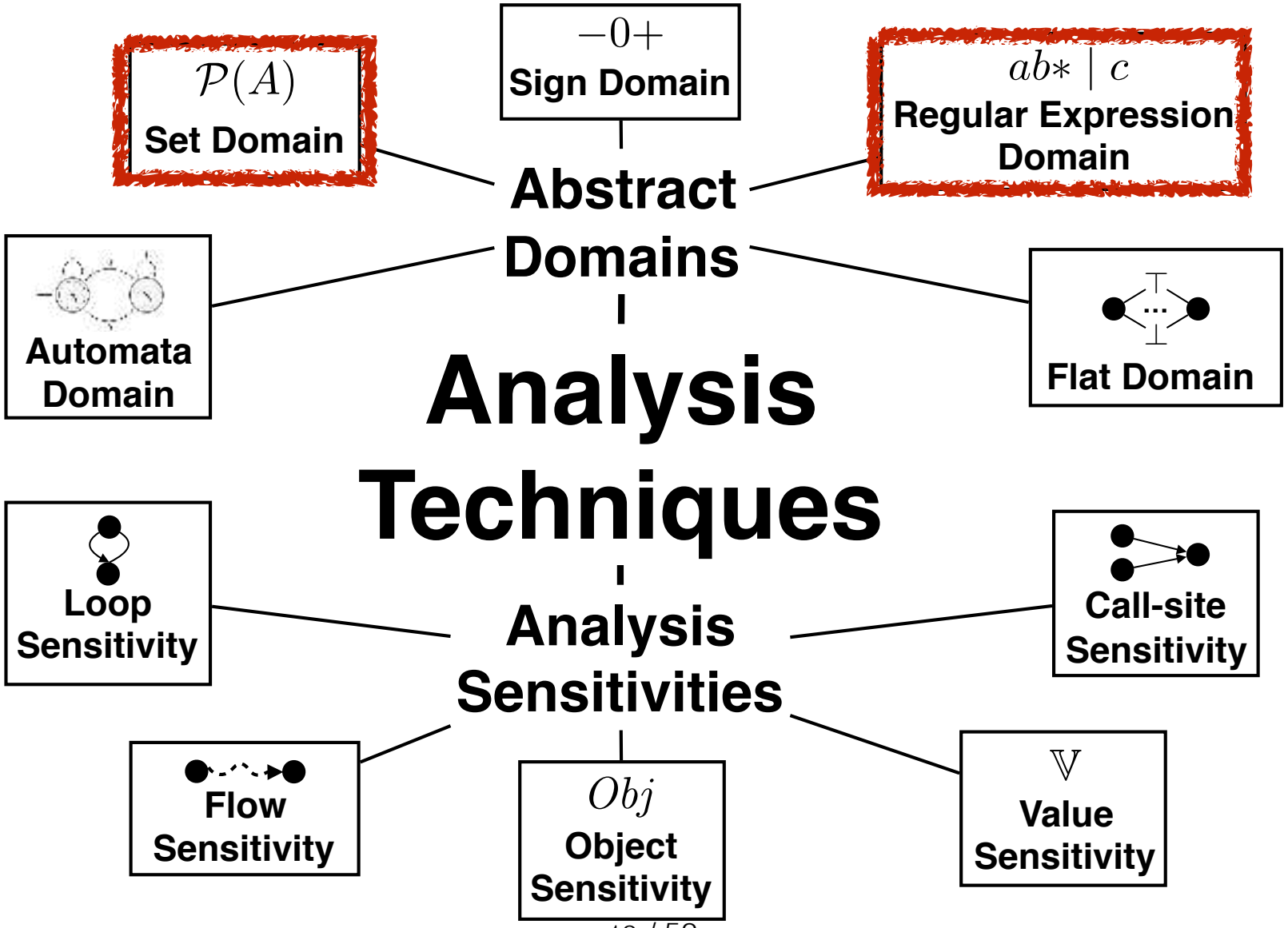
Pluggability



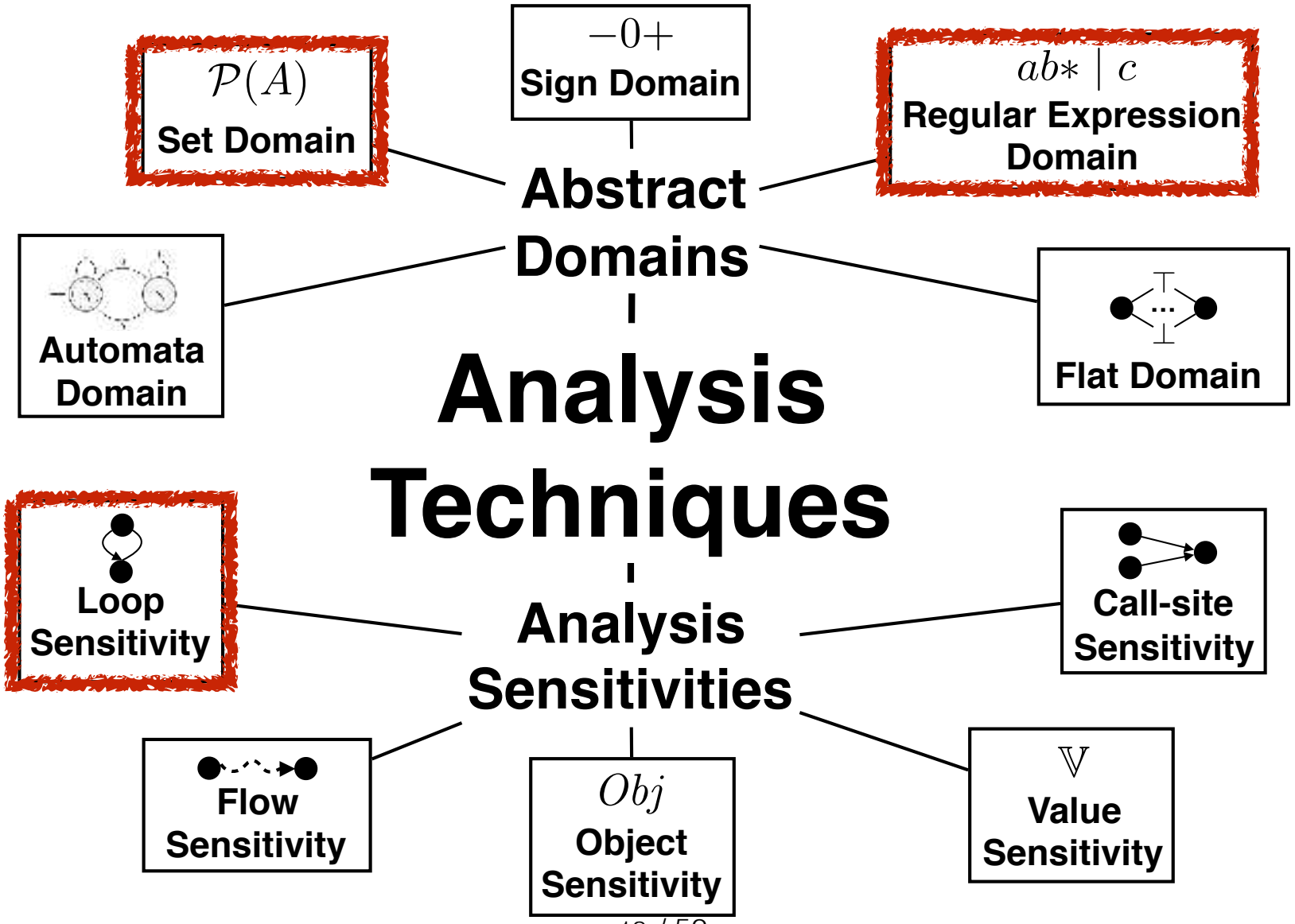
Pluggability



Pluggability



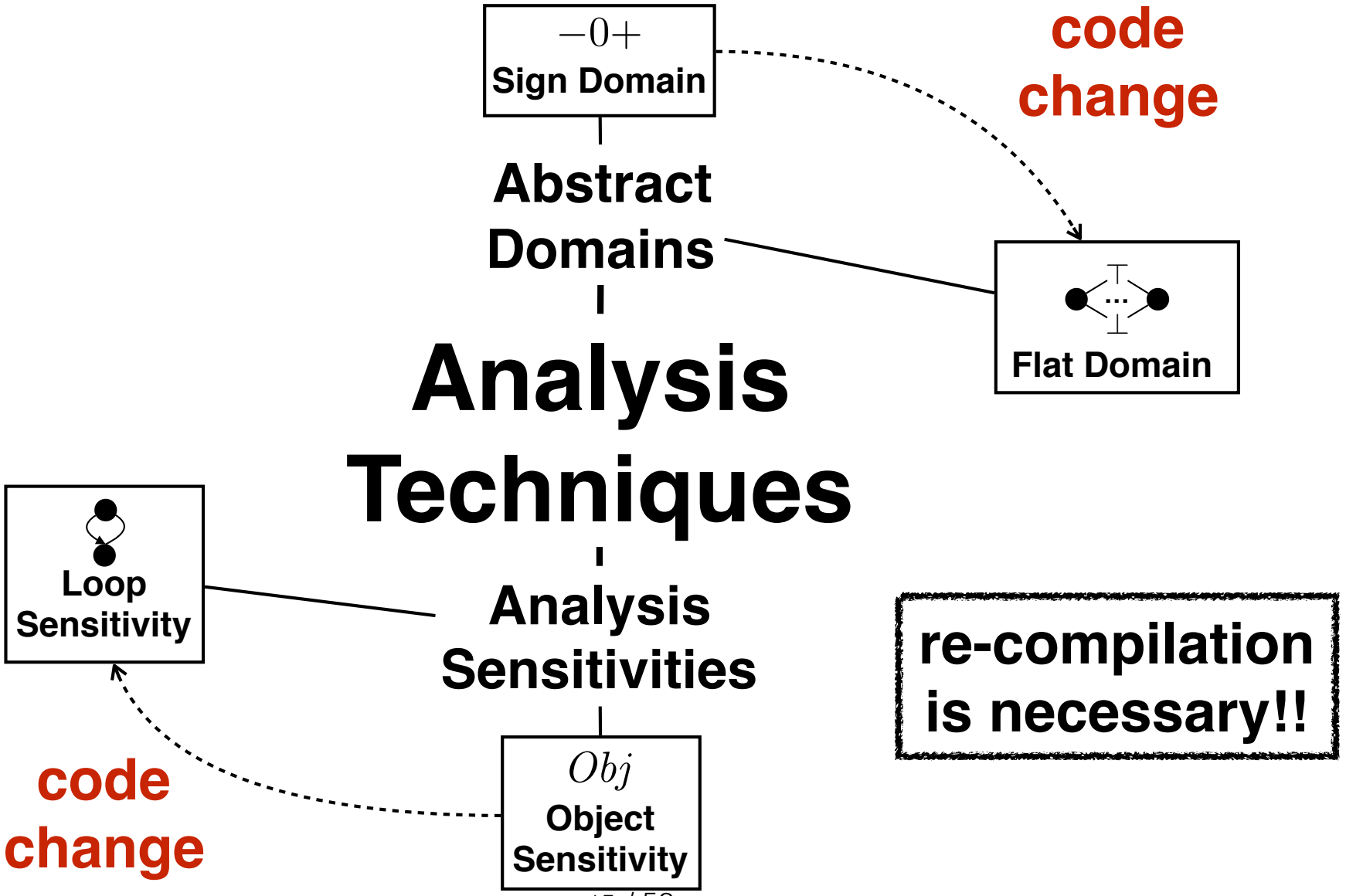
Pluggability



Pluggability



Pluggability



Pluggability

in SAFE 2.0

```
{
  "silent": false,
  "analyzer": {
    "number": "set",
    "string": "regular",
    "loopSens": 10,
  }
}
```

config1.json

Pluggability

in SAFE 2.0

```
{
  "silent": false,
  "analyzer": {
    "number": "set",
    "string": "regular",
    "loopSens": 10,
  }
}
```

config1.json

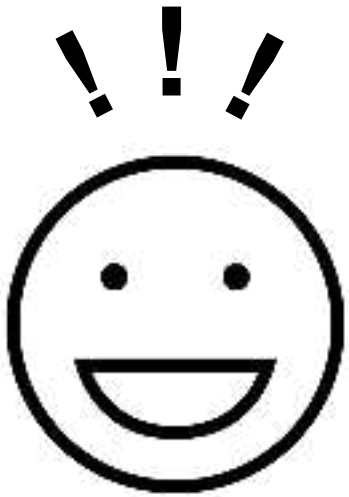


```
{
  "silent": false,
  "analyzer": {
    "number": "flat",
    "string": "regular",
    "loopSens": 10,
  }
}
```

config2.json

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Analysis Developers

Pluggability

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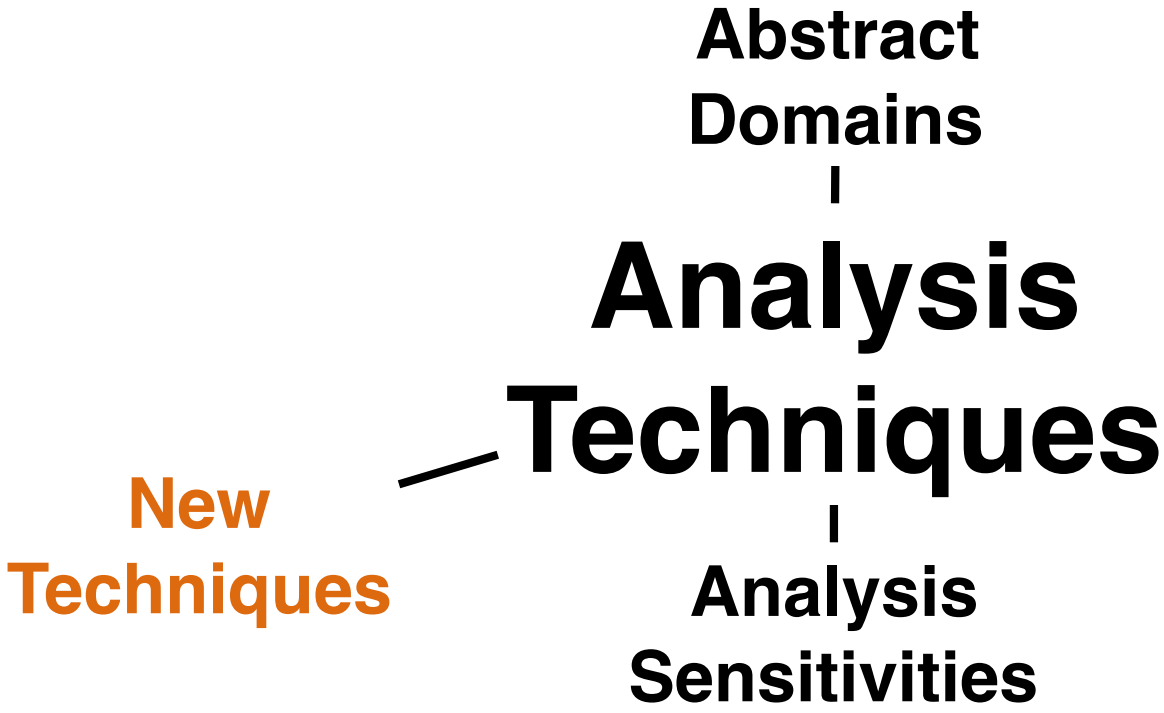
Extensibility

Abstract Domains

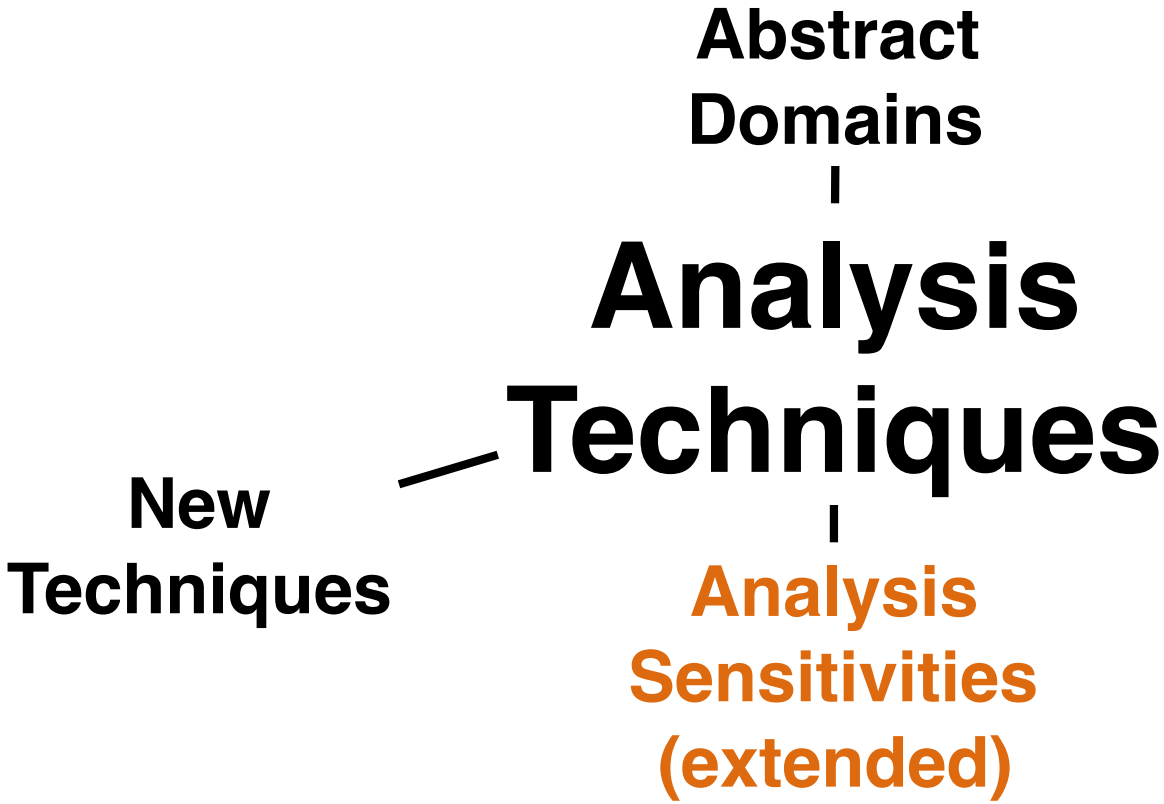
Analysis Techniques

Analysis Sensitivities

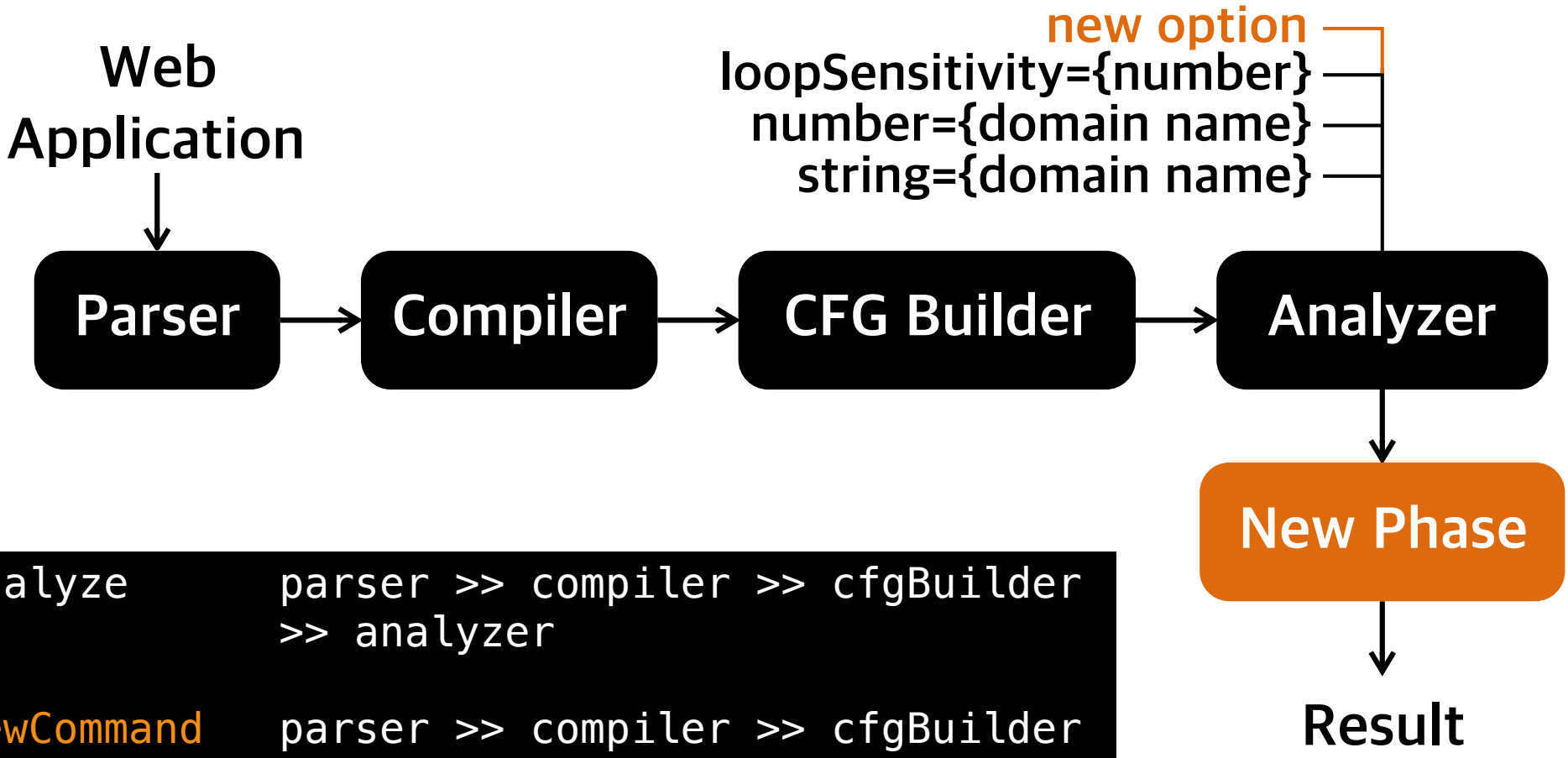
Extensibility



Extensibility



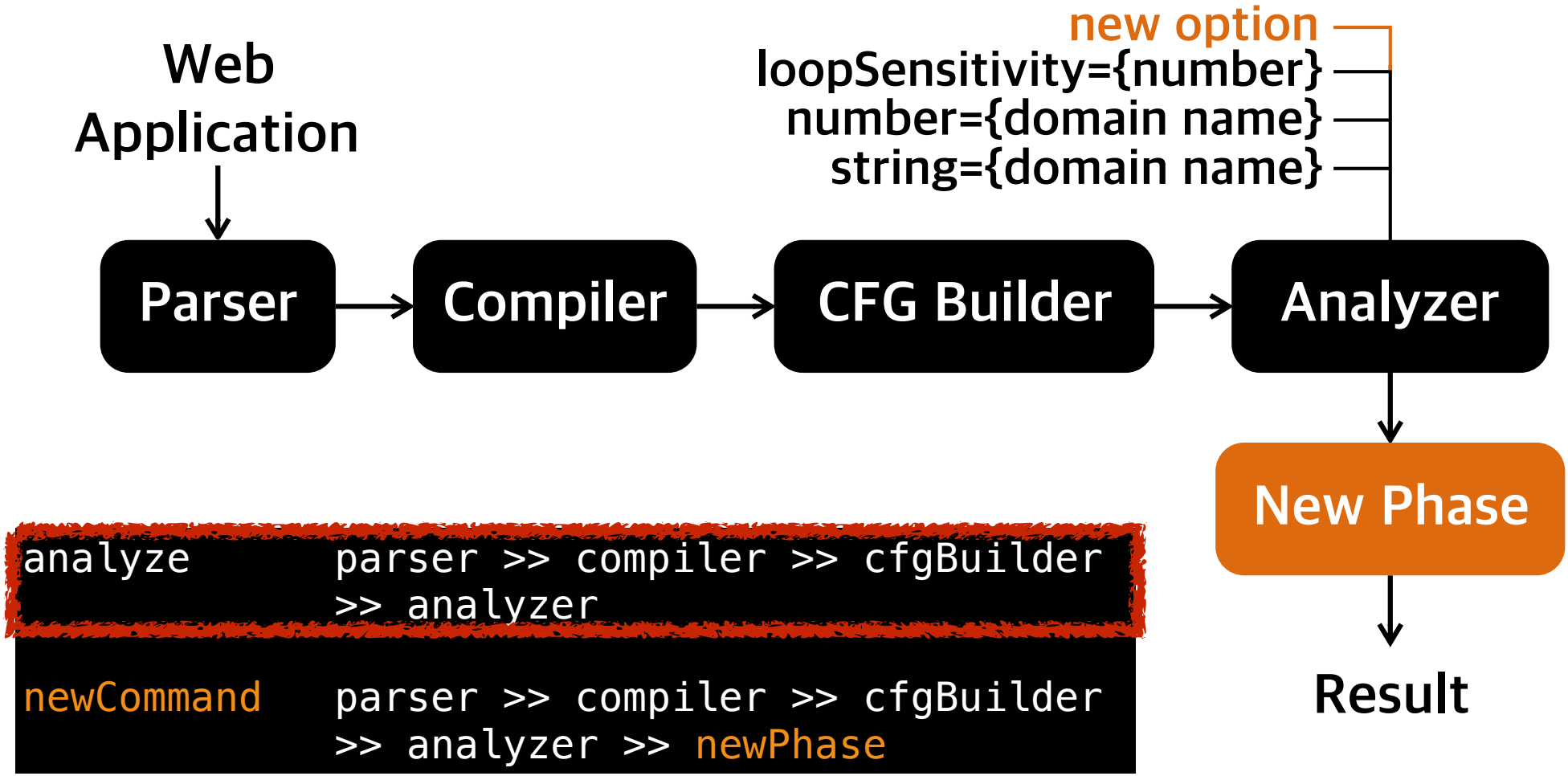
Extensibility




```
analyze       parser >> compiler >> cfgBuilder
              >> analyzer

newCommand    parser >> compiler >> cfgBuilder
              >> analyzer >> newPhase
```

Extensibility




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analyze       parser >> compiler >> cfgBuilder
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newCommand    parser >> compiler >> cfgBuilder
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Extensibility



Bug Detector

ASE'15

By New Phase

Extensibility




```
23 // BugDetect phase
24 case object BugDetector extends PhaseObj[
25   AnalysisResult, BugDetectConfig, BugReport
26 ] {
27   val name: String = "bugDetector"
28   val help: String = "Detect possible bugs in
29                       JavaScript source files."
30   val options: List[PhaseOption[BugDetectConfig]] = List(
31     ("silent", BoolOption(c => c.silent = true),
32       "messages during bug detection are muted.")
33   )
34
35   def apply(
36     in: AnalysisResult,
37     safeConfig: SafeConfig,
38     config: BugDetectConfig
39   ): Try[BugReport] = { ... }
```

Extensibility




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Extensibility




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Basic Info

Extensibility

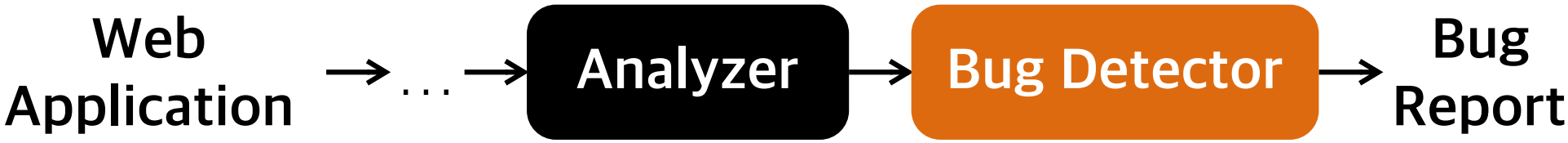



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```

Basic Info

Core functionality

Extensibility




```
136 ...
137 // androidCheck
138 case object CmdAndroidCheck extends CommandObj("androidCheck",
139   CmdBase >> AndroidCheck
140 )
141
142 // help
143 case object CmdHelp extends CommandObj("help", CmdBase >> Help)
144
145 // bugDetect
146 case object CmdBugDetect extends CommandObj("bugDetect",
147   CmdAnalyze >> BugDetector
148 )
```

bugDetect Command

Extensibility




```
$ safe bugDetect wikipedia.org.htm
wikipedia.org.htm:179:7-36:
  [Warning] The conditional expression "uiLang.match(new
            RegExp("^\w+", "")) === lang" is always false.
wikipedia.org.htm:192:5-18:
  [Warning] The property "chrome" of the object "window"
            is absent.
The command 'bugDetect' took 2547 ms.
$
```

False Alarm!!

Extensibility



Snapshot (Dynamic Info.)

ICSE'16

By New Option

Extensibility

Analyzer — new option

```
127       ...
128       ("jsModel", BoolOption(c => c.jsModel = true),
129         "analysis with JavaScript models."),
130       ("snapshot", StrOption((c, s) => c.snapshot = Some(s)),
131         "analysis with an initial heap generated from
132          a dynamic snapshot(*.json).")
133     )
134 }
135
136 // Analyze phase config
137 case class AnalyzeConfig(
138   var snapshot: Option[String] = None,
139   var silent: Boolean = false,
140   var console: Boolean = false,
141   ...
```

Extensibility

Analyzer — new option

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127       ...
128       ("jsModel", BoolOption(c => c.jsModel = true),
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Extensibility

Analyzer — new option

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Extensibility

Analyzer — new option

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141   ...
```

Extensibility

Analyzer — new option

```
52    ...
53    def addSnapshot(st: AbsState, snapshot: String): AbsState = {
54      val concreteHeap = Heap.parse(snapshot)
55      val abstractHeap = AbsHeap.alpha(concreteHeap)
56      AbsState(st.heap + abstractHeap, st.context)
57    }
58    ...
```

Parse

Abstract

Join

Extensibility

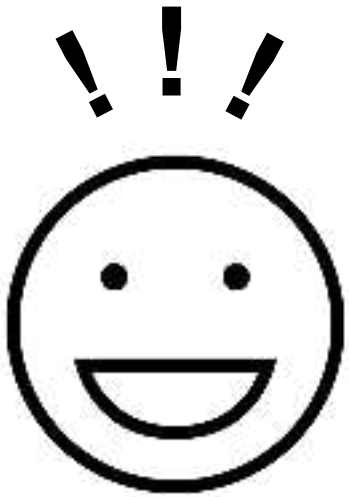


No False Alarm!!

```
$ safe bugDetect -analyzer:snapshot=chrome wikipedia.org.htm
wikipedia.org.htm:179:7-36:
  [Warning] The conditional expression “uiLang.match(new
            RegExp("^\w+", "")) === lang" is always false.
The command 'bugDetect' took 2168 ms.
$
```

SAFE 2.0

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Analysis Developers

Pluggability

Extensibility

Debuggability

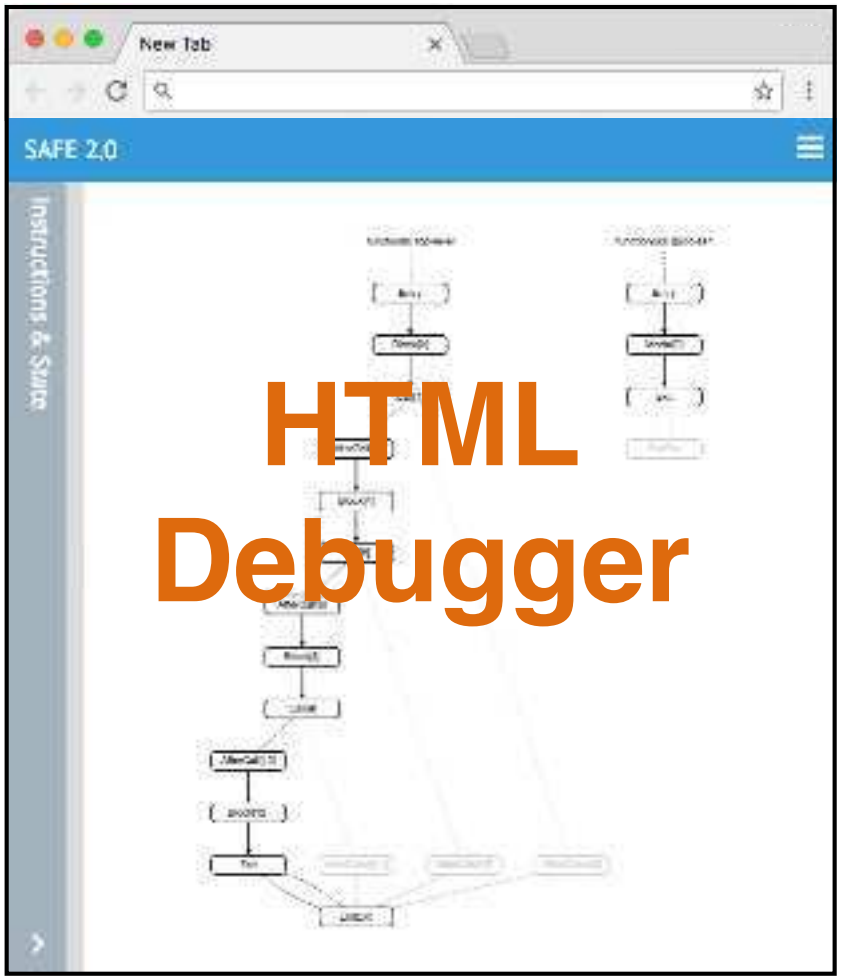
Debuggability

The official ECMAScript conformance suite

↑

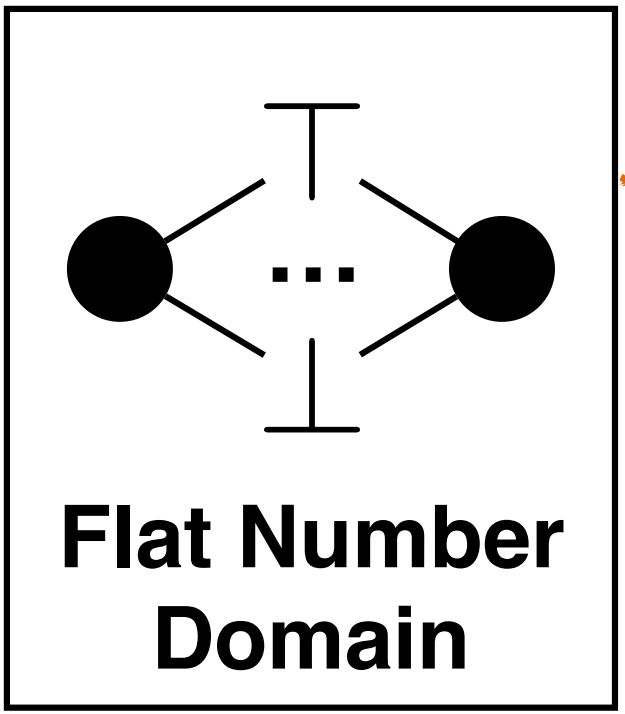
Test262

tests



HTML Debugger

Debuggability



SAFE 2.0

Debuggability

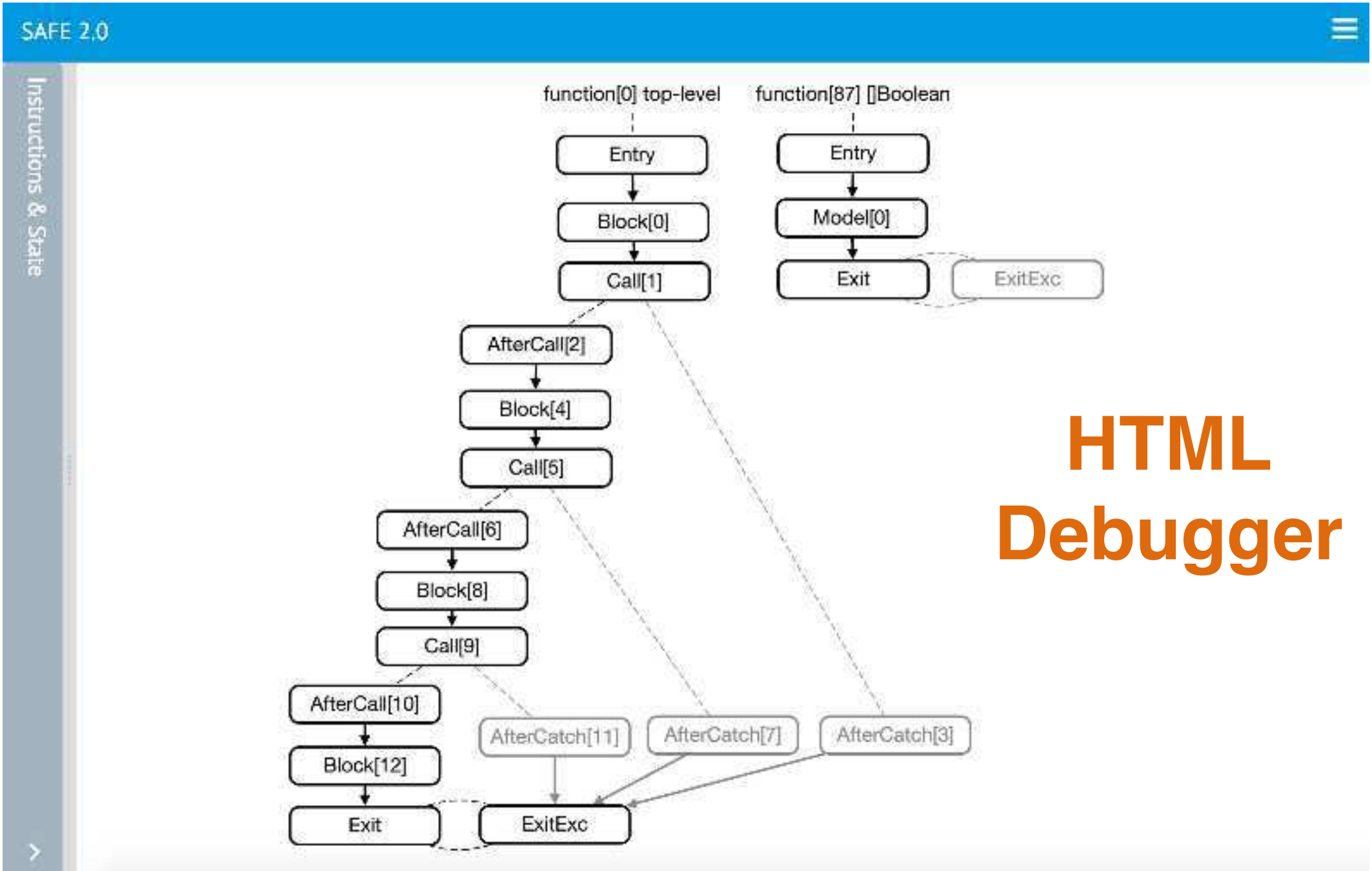
```
> test262Test
[info] Run completed in 14 minutes, 49 seconds.
[info] Total number of tests run: 5997
[info] Suites: completed 1, aborted 0
[info] Tests: succeeded 5983, failed 14, canceled 0, igno
red 0, pending 0
[info] *** 14 TESTS FAILED ***
[error] Failed tests:
[error]         kr.ac.kaist.safe.CoreTest
[error] (test:testOnly) sbt.TestsFailedException: Tests u
nsuccessful
[error] Total time: 896 s, completed Nov 16, 2016 12:45:0
2 AM
```

Test262 tests

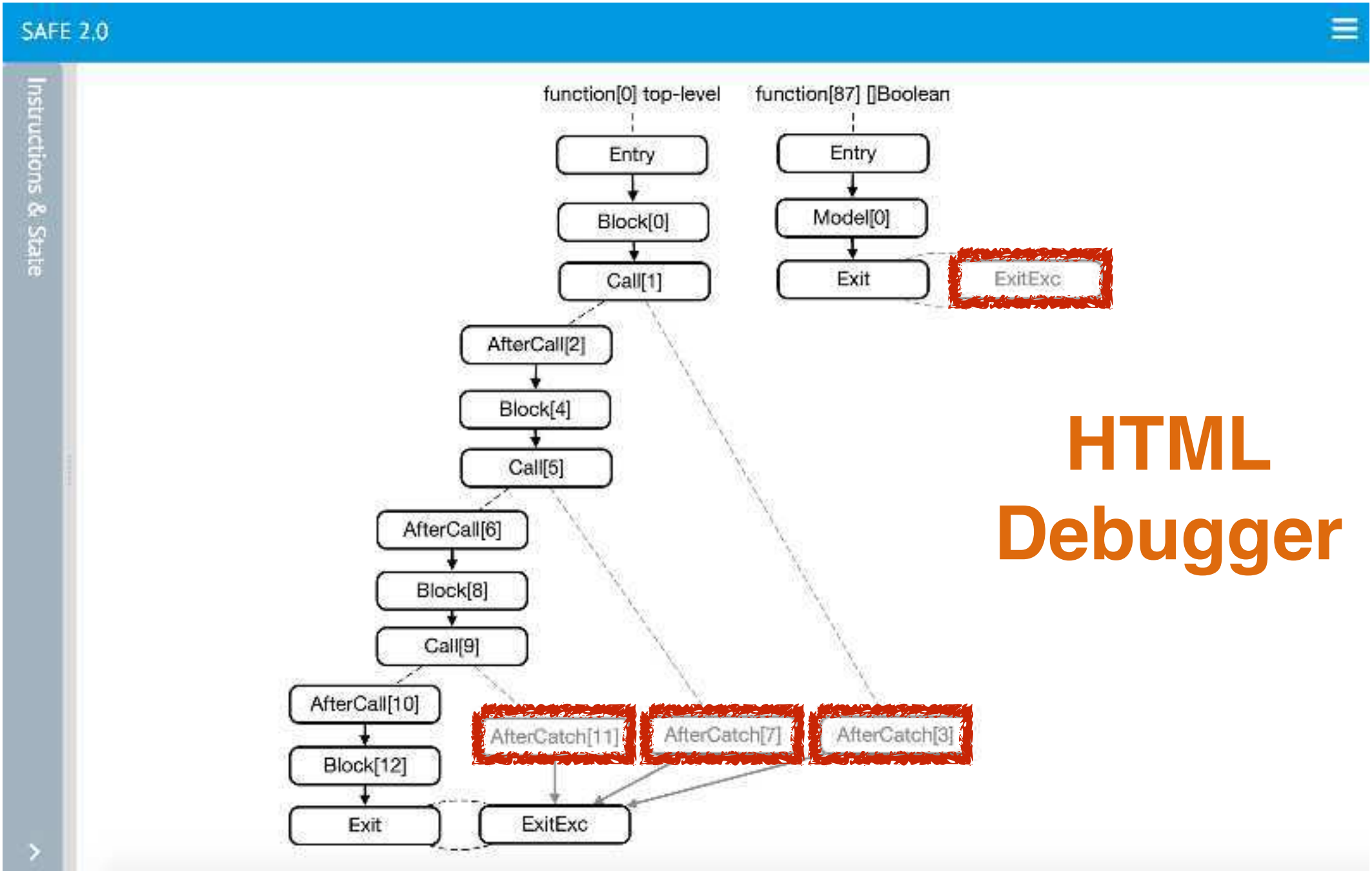
Debuggability

```
{
  var __result1 = Boolean(+ 0) !== false;
  var __expect1 = false;
}
{
  var __result2 = Boolean(- 0) !== false;
  var __expect2 = false;
}
{
  var __result3 = Boolean(Number.NaN) !== false;
  var __expect3 = false;
}
```

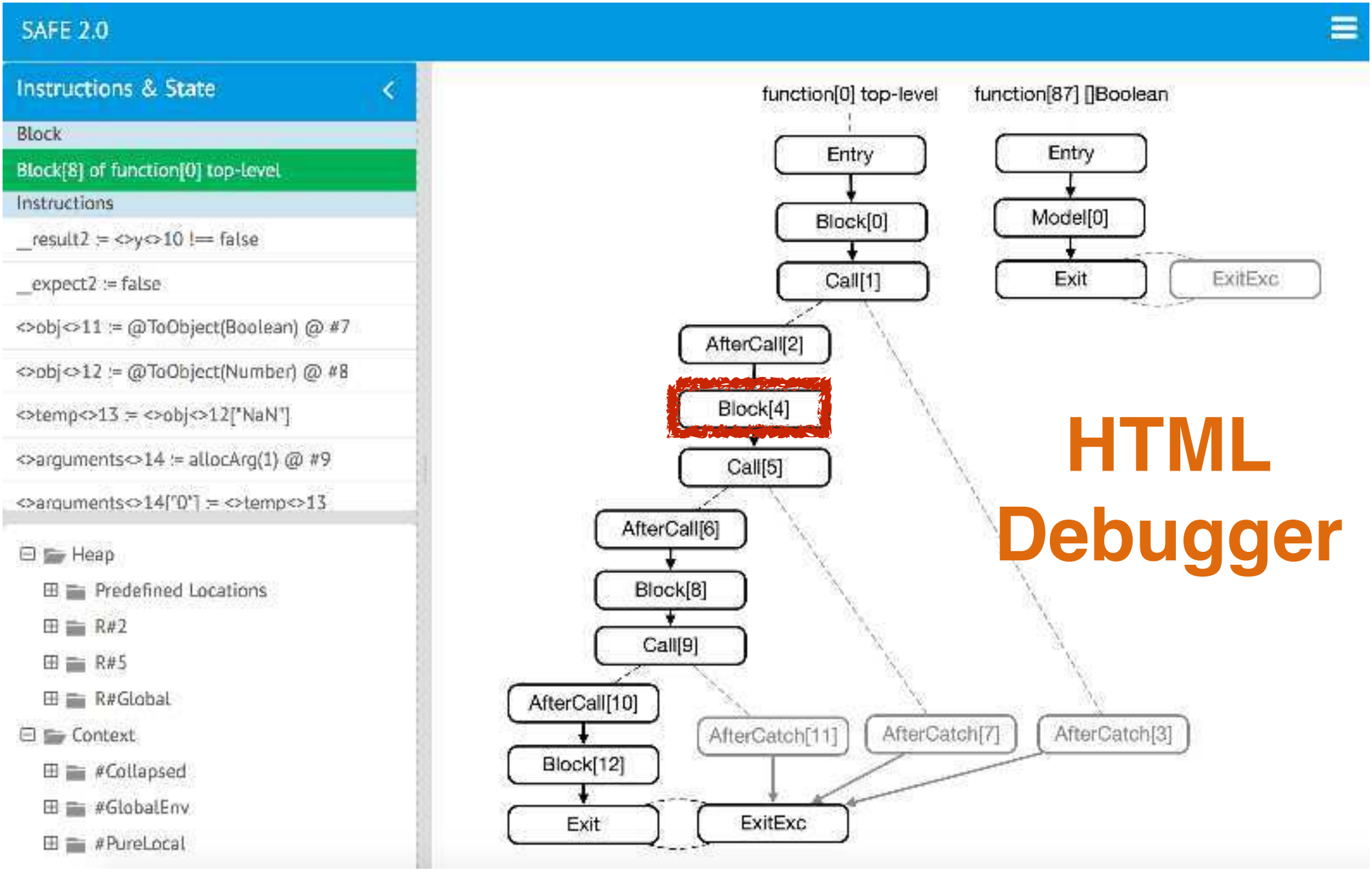
Debuggability



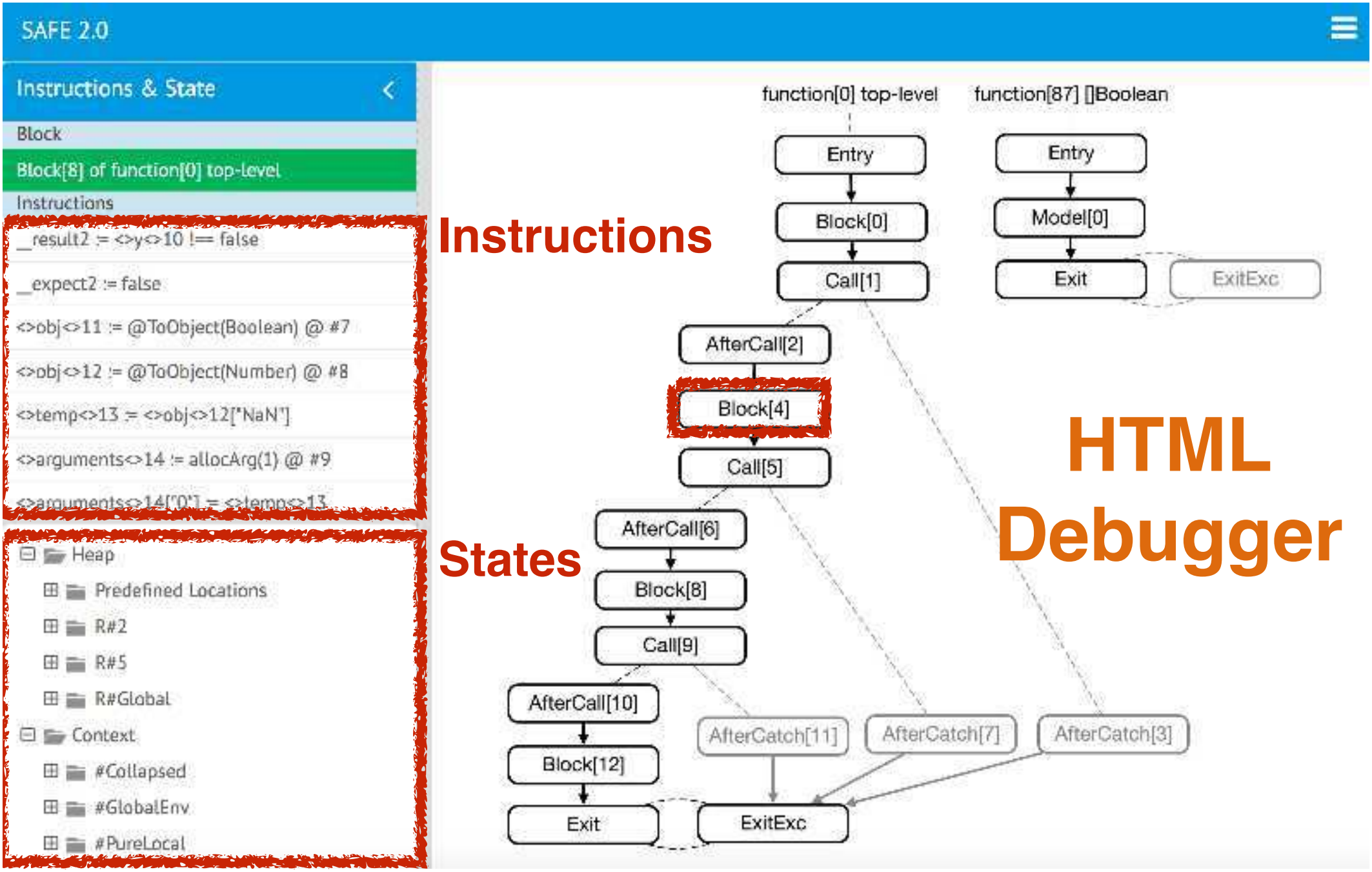
Debuggability



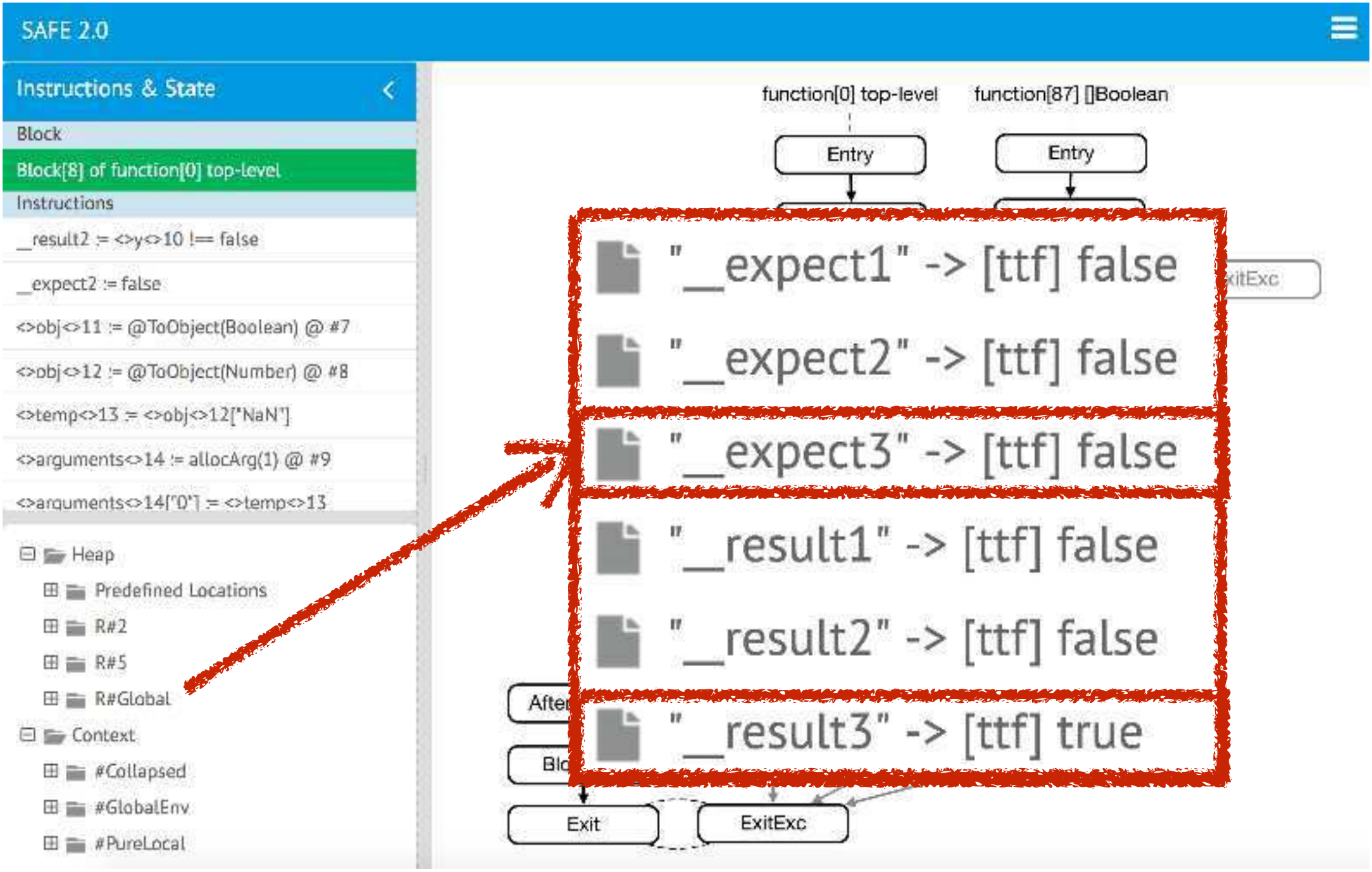
Debuggability



Debuggability



Debuggability



Debuggability

```
{
  var __result1 = Boolean(+ 0) !== false;
  var __expect1 = false;
}
{
  var __result2 = Boolean(- 0) !== false;
  var __expect2 = false;
}
{
  var __result3 = Boolean(Number.NaN) !== false;
  var __expect3 = false;
}
```

Debuggability

```
// 9.2 ToBoolean
def toAbsBoolean: AbsBool = this match {
  case Bot => AbsBool.Bot
  case Const(0.0) => AbsBool.False
  case Const(n) if n.isNaN => AbsBool.False
  case Const(_) => AbsBool.True
  case _ => AbsBool.Top
}
```

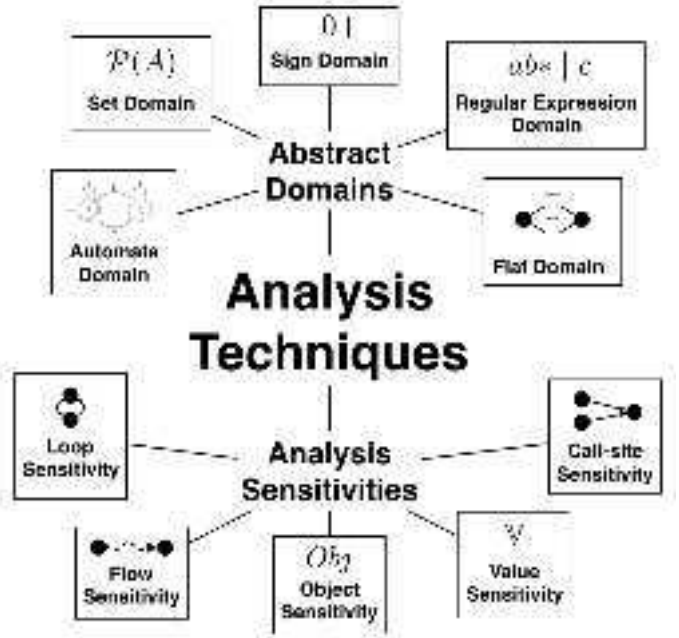
Flat Number Domain

Debuggability

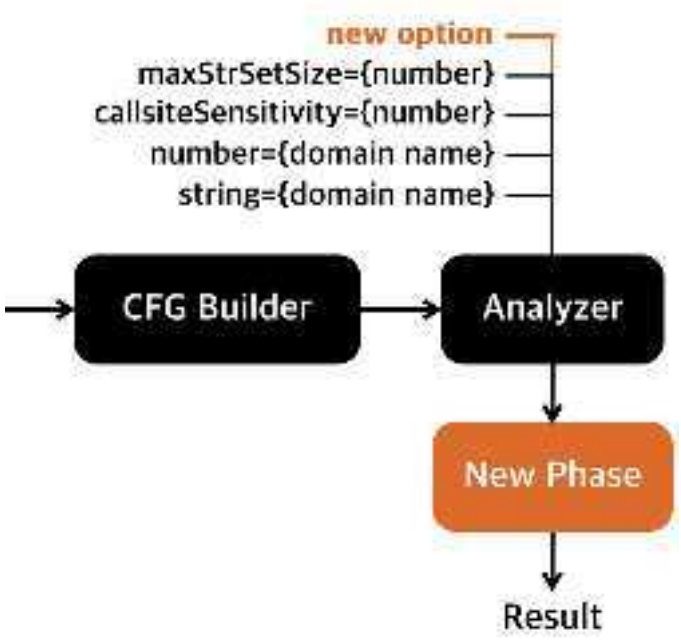
```
> test262Test
[info] Run completed in 17 minutes, 59 seconds.
[info] Total number of tests run: 5997
[info] Suites: completed 1, aborted 0
[info] Tests: succeeded 5997, failed 0, canceled 0, ignor
ed 0, pending 0
[info] All tests passed.
[success] Total time: 1082 s, completed Nov 16, 2016 1:39
:28 AM
```

Test262 tests

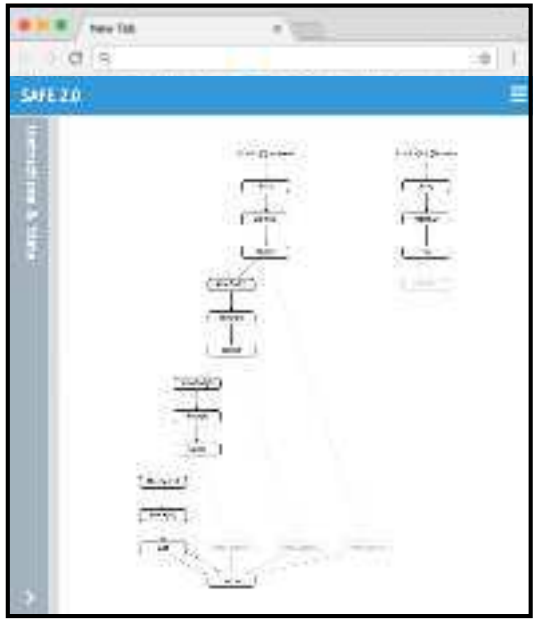
SAFE 2.0



Pluggability



Extensibility



Debuggability

https://github.com/sukyoung/safe